EXPERIMENT NO: 4

EXPLORING THE IP

AIM

To study the IP address configuration of a computer and explore basic IP related commands.

COMMANDS

1. ipconfig /all

Description: Displays the IP address, subnet mask, gateway and MAC address of the computer

```
IPv4 Address: 192.168.10.77
Subnet Mask : 255.255.255.0
Default Gateway: 192.168.10.1
```

2. ping

Description: Tests connectivity to another host

```
C:\>ping 192.168.10.1
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
```

3. nslookup

Description: Finds the IP address of a domain name

```
C:\>nslookup google.com
Name: google.com
Address: 142.250.193.78
```

4. tracert

Description: Shows the route taken by packets to a destination

```
C:\>tracert google.com
1  192.168.10.1   1 ms
2  10.10.0.1      4 ms
```

5. arp -a

Description: Displays the IP to MAC address mapping table

```
C:\>arp -a
192.168.10.1   3c-84-6a-7e-04-22
```

RESULT

The IP configuration of the computer was explored and basic IP related commands were executed successfully.